

Migration Validation & Risk Audit

Client: Compliance test
Migration: a to b
Audit Date: 2026-02-03
Auditor: Independent Migration Audit

Executive Summary

Final Verdict: NO-GO

- **Data Volume Checks:** FAIL
- **Aggregate Checks:** FAIL
- **Mapping Checks:** FAIL
- **Data Constraint Checks:** FAIL
- **Relationship Checks:** FAIL

Data Volume Checks

Check Name	Status	Message
Volume Check: users	FAIL	Row count difference exceeds tolerance for table 'users'. Source: 102, Target: 100, Difference: 1.96%.
Volume Check: orders	FAIL	Row count difference exceeds tolerance for table 'orders'. Source: 102, Target: 100, Difference: 1.96%.
Volume Check: order_items	FAIL	Row count difference exceeds tolerance for table 'order_items'. Source: 102, Target: 100, Difference: 1.96%.

Volume Check: products	FAIL	Row count difference exceeds tolerance for table 'products'. Source: 102, Target: 100, Difference: 1.96%.
------------------------	------	---

Aggregate Checks

Check Name	Status	Message
Sum Check: users - age	FAIL	Sum difference exceeds tolerance for column 'age' in table 'users'. Source: 3634, Target: 3559, Difference: 2.06%.
Average Check: users - age	WARN	Average difference within tolerance for column 'age' in table 'users'. Source: 35.627450980392155, Target: 35.59, Difference: 0.11%.
Max Check: users - age	PASS	Max matches exactly for column 'age' in table 'users'. Source and Target both have max 41.
Min Check: users - age	PASS	Min matches exactly for column 'age' in table 'users'. Source and Target both have min 30.
Variance Check: users - age	WARN	Variance difference within tolerance for column 'age' in table 'users'. Source: 9.364783537177246, Target: 9.436262626262627, Difference: 0.76%.
Sum Check: orders - amount	FAIL	Sum difference exceeds tolerance for column 'amount' in table 'orders'. Source: 20280.98, Target: 19875.98, Difference: 2.00%.

Average Check: orders - amount	WARN	Average difference within tolerance for column 'amount' in table 'orders'. Source: 198.83313725490197, Target: 198.75979999999998, Difference: 0.04%.
Max Check: orders - amount	PASS	Max matches exactly for column 'amount' in table 'orders'. Source and Target both have max 330.0.
Min Check: orders - amount	PASS	Min matches exactly for column 'amount' in table 'orders'. Source and Target both have min 50.0.
Variance Check: orders - amount	FAIL	Variance difference exceeds tolerance for column 'amount' in table 'orders'. Source: 6618.199209862164, Target: 6684.8302080404055, Difference: 1.01%.
Sum Check: orders - quantity	FAIL	Sum difference exceeds tolerance for column 'quantity' in table 'orders'. Source: 236, Target: 231, Difference: 2.12%.
Average Check: orders - quantity	WARN	Average difference within tolerance for column 'quantity' in table 'orders'. Source: 2.3137254901960786, Target: 2.31, Difference: 0.16%.
Max Check: orders - quantity	PASS	Max matches exactly for column 'quantity' in table 'orders'. Source and Target both have max 5.
Min Check: orders - quantity	PASS	Min matches exactly for column 'quantity' in table 'orders'. Source and Target both have min 1.

Variance Check: orders - quantity	FAIL	Variance difference exceeds tolerance for column 'quantity' in table 'orders'. Source: 1.7025820229081736, Target: 1.7312121212121216, Difference: 1.68%.
Sum Check: order_items - price	FAIL	Sum difference exceeds tolerance for column 'price' in table 'order_items'. Source: 11173.05, Target: 10933.05, Difference: 2.15%.
Average Check: order_items - price	WARN	Average difference within tolerance for column 'price' in table 'order_items'. Source: 109.53970588235293, Target: 109.33049999999999, Difference: 0.19%.
Max Check: order_items - price	PASS	Max matches exactly for column 'price' in table 'order_items'. Source and Target both have max 210.5.
Min Check: order_items - price	PASS	Min matches exactly for column 'price' in table 'order_items'. Source and Target both have min 50.0.
Variance Check: order_items - price	WARN	Variance difference within tolerance for column 'price' in table 'order_items'. Source: 816.7939771403611, Target: 818.413934090909, Difference: 0.20%.
Sum Check: order_items - quantity	FAIL	Sum difference exceeds tolerance for column 'quantity' in table 'order_items'. Source: 170, Target: 174, Difference: 2.35%.
Average Check: order_items - quantity	FAIL	Average difference exceeds tolerance for column 'quantity' in table 'order_items'. Source: 1.666666666666667, Target: 1.74, Difference: 4.40%.

Max Check: order_items - quantity	PASS	Max matches exactly for column 'quantity' in table 'order_items'. Source and Target both have max 4.
Min Check: order_items - quantity	WARN	Source min for column 'quantity' in table 'order_items' is zero.
Variance Check: order_items - quantity	FAIL	Variance difference exceeds tolerance for column 'quantity' in table 'order_items'. Source: 0.5808580858085808, Target: 0.47717171717171736, Difference: 17.85%.
Sum Check: products - price	WARN	Sum difference within tolerance for column 'price' in table 'products'. Source: 10867.98, Target: 10838.0, Difference: 0.28%.
Average Check: products - price	FAIL	Average difference exceeds tolerance for column 'price' in table 'products'. Source: 106.54882352941176, Target: 108.38, Difference: 1.72%.
Max Check: products - price	PASS	Max matches exactly for column 'price' in table 'products'. Source and Target both have max 999.99.
Min Check: products - price	PASS	Min matches exactly for column 'price' in table 'products'. Source and Target both have min 3.99.
Variance Check: products - price	FAIL	Variance difference exceeds tolerance for column 'price' in table 'products'. Source: 26291.31828771113, Target: 26649.21, Difference: 1.36%.
Sum Check: products - stock_quantity	FAIL	Sum difference exceeds tolerance for column 'stock_quantity' in table 'products'. Source: 29735, Target: 27935, Difference: 6.05%.

Average Check: products - stock_quantity	FAIL	Average difference exceeds tolerance for column 'stock_quantity' in table 'products'. Source: 291.51960784313724, Target: 279.35, Difference: 4.17%.
Max Check: products - stock_quantity	PASS	Max matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have max 2000.
Min Check: products - stock_quantity	PASS	Min matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have min 25.
Variance Check: products - stock_quantity	FAIL	Variance difference exceeds tolerance for column 'stock_quantity' in table 'products'. Source: 86184.05406717143, Target: 78477.60353535356, Difference: 8.94%.

Mapping Checks

Check Name	Status	Message
Mapping Check: users	FAIL	Mapping issues found in table 'users': Column 'status' in table 'users' has invalid values: ['FR' 'JP' 'CN' 'US' 'UK' 'CA' 'AU' 'DE'].
Mapping Check: users	FAIL	Mapping issues found in table 'users': Column 'country_code' in table 'users' has invalid values: [nan].
Mapping Check: orders	PASS	All mappings are valid for table 'orders'.
Mapping Check: orders	PASS	All mappings are valid for table 'orders'.
Mapping Check: order_items	PASS	All mappings are valid for table 'order_items'.

Mapping Check: products	PASS	All mappings are valid for table 'products'.
Mapping Check: products	PASS	All mappings are valid for table 'products'.

Data Constraint Checks

Check Name	Status	Message
Data Constraints Check: users.first_name	PASS	All data constraints are satisfied for table 'users'.
Data Constraints Check: users.last_name	PASS	All data constraints are satisfied for table 'users'.
Data Constraints Check: users.email	PASS	All data constraints are satisfied for table 'users'.
Data Constraints Check: users.date_of_birth	PASS	All data constraints are satisfied for table 'users'.
Data Constraints Check: users.date_onboarded	FAIL	Data constraint issues found in table 'users': Column 'date_onboarded' has 1 invalid date values out of 100 rows.
Data Constraints Check: orders.order_date	PASS	All data constraints are satisfied for table 'orders'.
Data Constraints Check: orders.amount	PASS	All data constraints are satisfied for table 'orders'.
Data Constraints Check: orders.user_id	PASS	All data constraints are satisfied for table 'orders'.
Data Constraints Check: order_items.order_id	PASS	All data constraints are satisfied for table 'order_items'.
Data Constraints Check: order_items.product_id	PASS	All data constraints are satisfied for table 'order_items'.
Data Constraints Check: order_items.quantity	PASS	All data constraints are satisfied for table 'order_items'.

Data Constraints Check: order_items.price	PASS	All data constraints are satisfied for table 'order_items'.
Data Constraints Check: products.product_name	PASS	All data constraints are satisfied for table 'products'.
Data Constraints Check: products.price	PASS	All data constraints are satisfied for table 'products'.
Data Constraints Check: products.stock_quantity	PASS	All data constraints are satisfied for table 'products'.

Relationship Checks

Check Name	Status	Message
Foreign Key Check: orders	PASS	All foreign key values in 'user_id' of child table 'orders' have matching primary keys in parent table.
Foreign Key check (Missing FK): orders	FAIL	Foreign key column 'order_id' not found in child table 'orders'.
Foreign Key Check: order_items	PASS	All foreign key values in 'order_id' of child table 'order_items' have matching primary keys in parent table.
Foreign Key Check: order_items	PASS	All foreign key values in 'product_id' of child table 'order_items' have matching primary keys in parent table.

Final Deployability Verdict

NO-GO